SatQuery AI - Multimodal Remote Sensing Evidence Report

Autonomous Agentic Execution & Evidence Provenance Trace

Query:	*"What changed between these two dates, and where did the change occur?"*
Task Category:	CHANGE_VQA
Specialist Tool:	Bi-Temporal Change Understanding & CDVQA Specialist
Reliability Score:	**86.5% (HIGH TRUST)**
Execution Time:	134.5 ms

1. Agent Synthesis & Visual-Language Answer

Bi-Temporal Change Analysis & CDVQA Output:
Bi-temporal change analysis identifies land-cover conversion between Observation T1 and T2.
1. Change Extent: 57.2% scene area modified (149,937 pixels).
2. Primary Hotspot Location: South-East portion of the scene tile.
3. Conversion Profile: Area expansion detected within primary land-cover category 'Inland wetlands (Marshes, Peat bogs)'.

Spatial Change Map generated with pixel-level deep feature differencing.

2. Auditable Agent Pipeline Execution Trace

• **[Perception Layer]** Format & CRS validated. Mode: Bi-Temporal Pair (T1 + T2) (2 image(s)). Status: PASSED
• **[Query Engine]** Classified intent: 'CHANGE_VQA'. Extracted entities: ['general_land_cover']. Task: 'change_vqa'
• **[Mission Planner]** Selected tool: 'Bi-Temporal Change Understanding & CDVQA Specialist' with params {'threshold_method': 'otsu', 'blur_kernel': [5, 5]}. Generated 7 node task-graph.
• **[Specialist Tool Execution]** Executed specialist tool 'Bi-Temporal Change Understanding & CDVQA Specialist' with model confidence 82.5%.
• **[Evidence Fusion]** Aligned 3 evidence streams. Temporal Alignment Confirmed: Spatial change map cross-validated across T1-T2 frames.
• **[Trust & Uncertainty]** Calculated Reliability Score: 86.5% (HIGH TRUST). Cross-model agreement: 83.4%.
• **[Discovery Engine]** Autonomous scan completed: Autonomous scan identified 2 'Beyond Your Query' discovery item(s).
• **[Answer Generator]** Final evidence-linked response synthesized in 134.5 ms.